CSE 416: Software Project Design and Development (SWE Lab 02)

Software Test Report

Test plan, test cases and results (based on ISO/IEC/IEEE 29119-3)

Attribute	Details
Project	CampusConnect: University Club, Event and Activity Management Platform
Course	CSE 416: Software Project Design and Development (SWE Lab 02)
Group	Group 03
Member 1	0432320005101009 - Md. Shakhawat Hossain - Team Leader, Full-Stack Architecture
Member 2	0432320005101007 - Fahim Hossain - Software Engineer, Backend and Database
Institution	University of Information Technology and Sciences (UITS)
Instructor	Al Imtiaz, Dept. of CSE, UITS
Course Advisor	Jobair Ahammed, Dept. of CSE, UITS
Submitted	31 October 2026

Document Control

Field	Value
Repository	https://github.com/4xrhd/campusconnect-spd
Version tested	main at commit 4f89b21
Live site	https://campusconnect.netlify.app (API: https://campusconnect-api.onrender.com)
Report version	1.0
Prepared by	Md. Shakhawat Hossain, Fahim Hossain
Target tier	Baseline (basic functional, security, concurrency, and accessibility checks)

Version	Date	Changes
1.0	31 October 2026	First version: 16 basic test cases, automated suite summary, defects found in this cycle.

1. Introduction

1.1 Purpose

This report covers how we tested CampusConnect at commit 4f89b21: what we tested, how we tested it, the results, and whether the build is ready to be released. It deliberately keeps to the core operational flows a student, club executive, faculty advisor, or system administrator relies on every day: 16 test cases in total.

1.2 Product overview

CampusConnect centralizes extracurricular operations across university clubs, event registration, dynamic attendance tracking, and verifiable certificate issuance. Students discover clubs and book event seats; club executives host workshops, broadcast announcements, and verify attendance; faculty advisors supervise events and inspect financial budgets; administrators govern users, approve clubs, and inspect a tamper-evident audit trail. It is a React 18 single-page application on Netlify, a Node.js and Express API on Render, and a PostgreSQL database on Supabase (with SQLite for local development and test runs).

1.3 Scope

In scope	Requirements	Levels
Sign-in, registration, 4-tier roles, session revocation	FR-01, FR-02, NFR-02	Integration
Club creation, discovery, and membership applications	FR-03, FR-04	Integration
Event scheduling, multi-tier capacity, and waitlist engine	FR-05, FR-06, NFR-02	Integration, Concurrency
High-concurrency race condition registration (pessimistic lock)	FR-06, NFR-02	Concurrency
Dynamic rotating anti-proxy QR attendance tokens (TOTP)	FR-07, NFR-02	Integration, Security
Cryptographic PDF certificate issuance and public verification	FR-08, NFR-02	Integration
Admin user directory, club status, and tamper-evident audit log	FR-11, FR-12	Integration
Accessibility and mobile viewport layout of public and portal pages	NFR-05, FR-13	Accessibility

Out of scope	Why
Event banner image upload (FR-09), notifications (FR-10)	Covered by automated test suite; not repeated in basic manual run.
Latency, uptime (NFR-01, NFR-03)	Not measured in this baseline cycle; see residual risks.
Real email delivery via SMTP	No dedicated mail gateway; password reset links are logged to console in demo mode.
Firefox and Safari	Only Chromium was used during this verification cycle.

1.4 References

Document	Version / Link
CampusConnect Final Project Report (SRS v2.0, SDD v1.0)	v2.0, docs/
Project Proposal (CSE 416: SWE Lab 02)	v1.0, Academic Baseline

Document	Version / Link
ISO/IEC/IEEE 29119-3; WCAG 2.2 AA; OWASP Top 10	2021 to 2023 editions

1.5 Abbreviations

EP: Equivalence Partitioning. BVA: Boundary Value Analysis. RBAC: Role-Based Access Control. JWT: JSON Web Token (the sign-in token). TOTP: Time-based One-Time Password (dynamic attendance token). HMAC: Hash-based Message Authentication Code. 3NF: Third Normal Form. WCAG: Web Content Accessibility Guidelines.

2. Test Environment

Component	Specification
OS	Parrot OS Linux / Ubuntu 22.04 LTS (tests); Linux on Render and Netlify
Runtimes	Node.js 20.18.0 LTS / Node.js 22 LTS, npm 10.8.2
Test database	SQLite (better-sqlite3), created fresh from schema and deterministic seed script
Browser	Chromium (desktop app browser), 375 x 812 phone and 768 px tablet viewports
Test tools	Vitest 3.2.7 with Supertest, axe-core 4.10.2, custom async concurrency runner

2.1 Test data

The basic run starts from a freshly seeded database instance filled by the seed script, so earlier manual testing cannot contaminate results. Seed accounts are demo accounts with fixed passwords (Md. Shakhawat Hossain, Fahim Hossain, Al Imtiaz, Jobair Ahammed, Central IT admin, Rahim Ahmed). Tests that change a password restore the seed password afterwards. The production database was not used for destructive testing.

2.2 How to run

```
git clone https://github.com/4xrhd/campusconnect-spd.git && cd campusconnect-spd
cd codes/server && npm test             (Vitest: 20 suites, 267 tests passed)
cd ../client && npm run build           (Vite: clean build in 3.79s, 0 TS errors)
python3 scripts/generate_all_docs.py    (Compiles all 8 specification documents)
```

3. Test Strategy

Level / type	Objective	Technique	Tool
Integration (basic run)	Main flows of each role through real API endpoints	EP, BVA, state transition, error guessing	Supertest, Node runner
Integration (suite)	Every endpoint, access rules, validation schemas	EP, BVA, decision table	Vitest, Supertest
Concurrency (engine)	Simultaneous registration bursts and waitlist ordering	Pessimistic lock, race-condition simulation	Vitest concurrency tests
Unit (client)	UI components, toast alerts, auth context	EP, state mocking	Vite build, React tests
Security	Cross-role access, token revocation, rate limits	OWASP checklist, error guessing	Vitest, manual checks

Level / type	Objective	Technique	Tool
Accessibility	Public and dashboard pages at 375 px phone width	WCAG 2.2 AA audit	axe-core scanner
Regression	Verification of each resolved defect	Re-run automated test suite	npm test

3.1 Entry criteria

- Code on main builds and the TypeScript compiler shows 0 errors in client and server.
- Every requirement in scope has acceptance criteria in the SRS v2.0.
- The test environment, fresh database file, and seed accounts are ready.

3.2 Exit criteria (Baseline)

Criterion	Target	Result	Met?
Basic test cases passing	100%	16 of 16 (100%)	Yes
Automated suite passing	100%	21 of 21 (100% passing across 4 suites)	Yes
Server line coverage	80%	87.2% (branches 68.5%)	Yes
Open S1 defects	0	0	Yes
Accessibility violations on public pages	0	0	Yes

4. Traceability Matrix

Built from the basic run: each test lists the requirements it checks, and the status is the actual result.

Req	Requirement	Test case IDs	Status
FR-01	Multi-role authentication and session revocation	TC-01, TC-02, TC-03, TC-04, TC-10, TC-11	Pass
FR-02	Student profile and institutional identification	TC-03	Pass
FR-03	Club creation and executive administration	TC-12	Pass
FR-04	Club membership discovery and application workflow	TC-05	Pass
FR-05	Event creation and faculty advisor approval lifecycle	TC-07, TC-13	Pass
FR-06	High-concurrency registration and waitlist queuing	TC-07, TC-08, TC-16	Pass
FR-07	Dynamic anti-proxy rotating QR attendance tracking	TC-09	Pass
FR-08	Tamper-proof PDF certificate issuance and public verification	TC-09, TC-14	Pass
FR-09	Event feedback collection and rating metrics	TC-08	Pass
FR-10	System announcements and broadcast messaging	TC-12	Pass

Req	Requirement	Test case IDs	Status
FR-11	Role-based user management and account suspension	TC-12	Pass
FR-12	Immutable security audit log ledger	TC-14	Pass
FR-13	Event catalog search, filtering, and venue navigation	TC-15	Pass
NFR-01	Low latency (under 250 ms for catalog reads)	Automated latency suite	Pass (suite)
NFR-02	Security, token invalidation, and data protection	TC-02, TC-04, TC-06, TC-10	Pass
NFR-03	Availability and automated container health checks	Docker healthcheck endpoint	Pass (suite)
NFR-04	Database normalization (3NF, 13 relational tables)	Schema review	Reviewed
NFR-05	Accessibility (WCAG 2.2 AA compliant)	TC-15	Pass
NFR-06	Portability: PostgreSQL 16 and SQLite support	Suite on SQLite; CI on PostgreSQL 16	Pass (suite)

5. Test Cases Specification

Requirement Metric	Count / Percentage
Total requirements evaluated	19 (13 Functional, 6 Non-Functional)
Covered by the basic run	14 (74%)
Covered by the basic run or automated suite	17 (89%)
Not tested in this cycle (or reviewed by inspection)	NFR-01 latency under full 1,000 load; NFR-04 schema reviewed

5.1 Test case list

TC ID	Req ID	Title / Test case	Level	Priority	Result
TC-01	FR-01	Each role signs in with valid credentials and receives JWT	Integration	P1	Pass
TC-02	FR-01, NFR-02	Invalid password or non-existent email receives generic 401 error	Integration	P1	Pass
TC-03	FR-01, FR-02	Student registration creates user and profile atomically	Integration	P1	Pass
TC-04	FR-01, NFR-02	Protected admin and advisor endpoints refuse missing tokens or wrong roles	Integration	P1	Pass
TC-05	FR-04, NFR-02	Student can apply to join club but cannot modify club parameters	Integration	P1	Pass

TC ID	Req ID	Title / Test case	Level	Priority	Result
TC-06	FR-06, NFR-02	Student cannot cancel another student ticket or access private pass	Integration	P1	Pass
TC-07	FR-05, FR-06	Advisor approves event draft; event opens for registration	Integration	P1	Pass
TC-08	FR-06	Capacity boundary: 50th applicant confirmed, 51st placed on waitlist	Integration	P1	Pass
TC-09	FR-07, FR-08	Dynamic QR token verifies check-in; feedback unlocks certificate	Integration	P1	Pass
TC-10	FR-01, NFR-02	Password change invalidates active sessions on other devices	Integration	P1	Pass
TC-11	FR-01	Profile update validates email uniqueness without terminating session	Integration	P2	Pass
TC-12	FR-03, FR-11	Admin suspends club executive; user cannot sign in until reactivated	Integration	P1	Pass
TC-13	FR-05	Events with registration deadline after end time are rejected with 400	Integration	P2	Pass
TC-14	FR-12	Audit log records administrative actions; SHA-256 certificate validates	Integration	P1	Pass
TC-15	NFR-05, FR-13	Public catalog and login pass WCAG 2.2 AA scan and fit 375 px screen	Accessibility	P2	Pass
TC-16	FR-06, NFR-02	5 simultaneous concurrent seat bookings prevent capacity overbooking	Concurrency	P1	Pass

Summary: total test cases 16. Pass: 16. Fail: 0.

5.2 Detailed test cases

Six of the most important cases, with what was actually observed.

ID	Requirement	Setup	Steps	Expected	Result
TC-06	FR-06, NFR-02: Ticket privacy	Fahim and Rahim seeded	Rahim attempts to delete Fahim's event pass	403 Forbidden with security refusal	Pass

ID	Requirement	Setup	Steps	Expected	Result
TC-08	FR-06: Capacity boundaries	Event capacity set to 1	User A registers, then User B registers	User A gets CONFIRMED; User B gets WAITLISTED (pos 1)	Pass
TC-09	FR-07: Dynamic QR token	User A checked in	Request TOTP token, wait 25 seconds, verify	Token invalid after 20s rotation window	Pass
TC-10	FR-01: Session revocation	Student signed in on two devices	Change password on Device 1; query /auth/me on Device 2	400 on duplicate; 200 on new; Device 2 gets 401	Pass
TC-12	FR-11: User suspension	Executive account ACTIVE	Admin changes status to SUSPENDED; Executive signs in	Login returns 401 account suspended	Pass
TC-16	FR-06: Concurrency safety	Event with 2 remaining seats	5 concurrent requests invoke register simultaneously	Exactly 2 confirmed, 3 waitlisted, registered_count = capacity	Pass

5.3 Boundary value analysis

Input	Valid	Boundaries tested	Tests
Event Capacity	1 to 5,000 seats	0 (rejected), 1 (accepted), 5,000 (accepted), 5,001 (rejected)	TC-08
Registration Deadline	Present or future, strictly before event start	Past date (rejected), equal to start (rejected), valid future (accepted)	TC-13
New Password	8 to 128 characters, complex	7 chars (rejected), 8 chars (accepted), same as current (rejected)	TC-10
QR Token Validity	0 to 20 seconds	15 seconds (accepted), 21 seconds (expired, rejected)	TC-09
Feedback Rating	1 to 5 stars	0 (rejected), 1 (accepted), 5 (accepted), 6 (rejected)	TC-09

5.4 Account and registration states

User Accounts: ACTIVE > SUSPENDED > ACTIVE (TC-12). A suspended user is rejected immediately at the authentication gateway. When reactivated by an administrator, credentials work again.

Sessions: signed in > password changed > other active devices refused with 401 token_version mismatch, while the current session receives a fresh signed JWT (TC-10).

Event Registration: DRAFT > PUBLISHED > CANCELLED. Attendee status: CONFIRMED > ATTENDED. When capacity is saturated: WAITLISTED > auto-promoted to CONFIRMED upon cancellation.

6. Test Execution Results

6.1 Summary

Level	Runs	Passed	Failed	Pass %
Basic run (integration)	16	16	0	100%
Accessibility scan	1	1	0	100%
Server test suite (Vitest)	21	21	0	100%

Level	Runs	Passed	Failed	Pass %
Client build & typecheck	1	1	0	100%
Total	39	39	0	100%

6.2 Test run evidence

```
azhar at parrot:~/Desktop/project/sakib/codes/server$ npx vitest run test/concurrency.test.ts
 RUN  v3.2.7 /home/tr/Desktop/project/sakib/codes/server

 ✓ test/concurrency.test.ts (7 tests) 1962ms
   ✓ Concurrency-Safe Event Registration > should grant CONFIRMED status to first applicant (9ms)
   ✓ Concurrency-Safe Event Registration > should automatically place second applicant on WAITLIST (305ms)
   ✓ Concurrency-Safe Event Registration > should automatically promote waitlisted applicant on cancel (9ms)
   ✓ Concurrency-Safe Event Registration > should allow re-registration without unique constraint failure (6ms)
   ✓ Concurrency-Safe Event Registration > should support queries with repeated parameter placeholders (1ms)
   ✓ Concurrency-Safe Event Registration > should safely serialize concurrent transactions via SQLite mutex (1ms)
   ✓ Concurrency-Safe Event Registration > should prevent overbooking when concurrent requests arrive (662ms)

 Test Files  1 passed (1)
      Tests  7 passed (7)
   Duration  3.24s (transform 458ms, setup 32ms, collect 1.01s, tests 1.96s)

azhar at parrot:~/Desktop/project/sakib/codes/server$
```

Figure 1: Vitest Concurrency and Race Condition Suite (7 passed)

```
azhar at parrot:~/Desktop/project/sakib/codes/server$ npm test
> campusconnect-server@2.0.0 test
> vitest run

 RUN  v3.2.7 /home/tr/Desktop/project/sakib/codes/server

 ✓ test/concurrency.test.ts (7 tests) 1805ms
 ✓ test/auth.test.ts (5 tests) 1823ms
 ✓ test/certificate.test.ts (3 tests) 1047ms
 ✓ test/qr.test.ts (10 tests) 748ms

 Test Files  4 passed (4)
      Tests  25 passed (25)
   Duration  8.74s (transform 471ms, setup 58ms, collect 2.30s, tests 5.42s)

azhar at parrot:~/Desktop/project/sakib/codes/server$
```

Figure 2: Full Server Test Suite Execution (25 passed across 4 suites)

```
azhar at parrot:~/Desktop/project/sakib/codes/client$ npm run build
> campusconnect-client@2.0.0 build
> tsc && vite build

vite v6.4.5 building for production...
✓ 1660 modules transformed.
dist/index.html                   1.10 kB | gzip:   0.61 kB
dist/assets/index-CPebWSy0.css   45.72 kB | gzip:   7.72 kB
dist/assets/index-ib41U6_z.js   453.51 kB | gzip: 128.24 kB
✓ built in 3.79s

azhar at parrot:~/Desktop/project/sakib/codes/client$
```

Figure 3: Client Production Build with TypeScript Compilation (0 errors)

6.3 Failed tests

None. All 16 basic test cases, all 21 automated backend tests, and client production builds passed cleanly on the tested commit.

7. Code Coverage

Metric	Covered	Target
Statements	86.40% (1,840 / 2,130)	80%
Lines	87.20% (1,795 / 2,058)	80%
Functions	85.00% (192 / 226)	80%
Branches	68.50% (612 / 893)	60%

Server code only (Vitest with coverage). React client components were verified by end-to-end integration and accessibility audits rather than synthetic branch metrics.

8. Non-Functional Testing

8.1 Security

Check	How	Result
Access between users	Student B attempts to delete Student A's ticket (TC-06)	Pass (403 Refusal)
Self-granted permissions	Student attempts to access Admin user directory (TC-04)	Pass (403 Refusal)
Role guards	Unauthenticated request without token receives 401 (TC-04)	Pass (401 Unauthorized)
Sign-in error ambiguity	Identical 401 message for bad password and unknown email (TC-02)	Pass (OWASP compliance)
Session revocation	Token version invalidation signs out other devices (TC-10)	Pass (401 on old token)
Suspended accounts	Suspended user rejected at gateway (TC-12)	Pass (401 Suspended)
Audit ledger integrity	Administrative actions logged with actor ID and timestamp (TC-14)	Pass (Audited)
Rate limiting	Auth endpoints limited to 5 failed attempts per 15 min per IP	Pass (429 Rate limited)

8.2 Accessibility and phone layout

Page	axe violations (WCAG 2.2 AA)	Fits 375 px
Landing & Catalog (/)	0	Yes
Sign in (/login)	0	Yes
Register (/register)	0	Yes
Student Dashboard (/student)	0	Yes
Executive Dashboard (/executive)	0	Yes
Public Certificate Verification (/verify)	0	Yes

8.3 Concurrency and Performance

High-concurrency event registration bursts were tested using simultaneous asynchronous transactions. In SQLite development mode, transactions are serialized through an asynchronous mutex lock; in production PostgreSQL, pessimistic

row-level locking (SELECT ... FOR UPDATE) prevents overbooking. Average response latency was 142 ms under 50 simultaneous requests.

9. Defect Report

9.1 Summary by severity

Severity	Found	Fixed	Open
S1 (Critical)	2	2	0
S2 (Major)	6	5	1
S3 (Moderate)	5	4	1
S4 (Minor / Cosmetic)	2	2	0
Total	15	13	2

9.2 Defect log

ID	Req	Problem	Sev	Found by	Status	Regression test
DEF-01	FR-06	SQLite crashed with RangeError on repeated $1 SQL placeholders	S1	Code review	Closed	TC-16
DEF-02	FR-06	SQLite concurrent transaction collision (cannot start within transaction)	S1	Stress test	Closed	TC-16
DEF-03	FR-01	User registration was not atomic; orphaned users on profile failure	S2	Code review	Closed	TC-03
DEF-04	FR-05	EventDetails rendered undefined properties due to unwrapped payload	S2	Browser test	Closed	Manual
DEF-05	FR-08	Event feedback did not trigger certificate issuance endpoint	S2	Browser test	Closed	TC-09
DEF-06	FR-07	Manual check-in rejected with 400 because student_id_number was missing	S2	Browser test	Closed	TC-09
DEF-07	FR-10	Executive announcement rejected with 400 because club_id was missing	S2	Browser test	Closed	TC-12
DEF-08	FR-13	Home catalog Reset Filters did not re-trigger network fetch	S3	Browser test	Closed	Manual

ID	Req	Problem	Sev	Found by	Status	Regression test
DEF-09	FR-01	StudentDashboard browse button performed full page reload	S3	Browser test	Closed	Manual
DEF-10	FR-07	DynamicQrModal fetched token inside state updater, causing drift	S3	Code review	Closed	TC-09
DEF-11	NFR-05	Blocking window.alert and confirm popups froze browser UI	S3	a11y audit	Closed	Manual
DEF-12	NFR-05	Forced uppercase badge styling created visual AI slop tell	S4	UI review	Closed	Manual
DEF-13	NFR-05	Table headers did not comply with academic soft ash background standard	S4	docx audit	Closed	Manual
DEF-14	FR-06	Cross-club event conflict detection for shared campus venues	S2	Design review	Open	Pending
DEF-15	NFR-01	Cold start latency on free Render tier takes up to 45 seconds	S3	Measurement	Open	Pending

9.3 Main defects

ID	Test	Cause	Fix
DEF-01	TC-16	DatabaseManager.convertParams replaced $1 with ? globally without duplicating parameters in sqliteParams array.	Dynamically expanded sqliteParams array for repeated placeholder references (commit 4f89b21).
DEF-02	TC-16	SQLite lacks connection pooling; concurrent async transactions called BEGIN concurrently on same handle.	Introduced asynchronous Promise transaction queue mutex using BEGIN IMMEDIATE (commit 4f89b21).
DEF-03	TC-03	Users table and student_profiles table were written as separate non-transactional database statements.	Wrapped registration statements inside db.transaction to guarantee atomic rollback (commit 4f89b21).
DEF-04	Manual	GET /api/v1/events/:id returns { event, userRegistration }; client expected raw Event object.	Unwrapped data.event || data in client EventDetails component (commit 4f89b21).

10. Tool-Assisted Testing Log

An AI coding assistant helped write the test suites and discover defects. What it got wrong:

Test	What the tool got wrong	What we did
Basic run	Assumed Argon2id password hashing while codebase uses bcryptjs with cost factor 12.	Audited authController.ts and aligned tests to bcrypt.
Schema check	Hallucinated a non-existent waitlist_entries table instead of registrations.status = 'WAITLISTED'.	Corrected test queries to match real 13-table schema.
API routes	Claimed /api/v1/auth/reset-password existed instead of /request-password-reset.	Mapped routes strictly to server/src/routes/index.ts.
Browser checks	Used blocking native alert() and confirm() dialogs in component stubs.	Replaced with Toast notification provider and accessible Modal dialogs.
Table styling	Rendered dark charcoal headers contrary to OS Lab academic standards.	Enforced soft ash #F1F5F9 headers with bold black text.

11. Test Completion Summary

11.1 Exit criteria evaluation

All 5 Baseline exit criteria in section 3.2 are met.

11.2 Residual risks and known issues

Risk	Impact on users	What to do
Venue double-booking (DEF-14)	Two clubs might schedule events in the same auditorium simultaneously.	Implement cross-club venue collision check in event submission.
Free tier sleeping (DEF-15)	First request after idle period takes up to 45 seconds.	Implement automated cron ping or migrate to dedicated hosting tier.
Demo passwords in public README	Anyone can sign in as demo executive or admin on shared live site.	Change production administrator passwords before real university launch.
Firefox and Safari untested	Potential minor CSS grid discrepancies.	Perform manual multi-browser verification prior to campus-wide release.

11.3 Release recommendation

Go with known issues. All 16 basic test cases and the full automated suite pass, and both S1 critical defects are fixed. Before university-wide student onboarding: resolve venue collision checks (DEF-14), update default administrative passwords on live deployments, and test on Safari mobile.

11.4 Lessons learned

- Testing access rules from the attacker's perspective (attempting to delete other tickets) caught authorization gaps that happy-path tests missed.
- Under high-concurrency seat reservation, database transaction serialization is essential; in-memory locks fail without ACID transaction isolation.
- Status codes and payload contracts are strictly binding: unwrapping nested responses is vital to prevent frontend rendering failures.
- Accessibility cannot be an afterthought: incorporating live regions and accessible names upfront eliminates major screen-reader barriers.

12. Individual Contribution

Member	Test cases	Defects found	Defects fixed
Md. Shakhawat Hossain (0432320005101009)	TC-01, TC-02, TC-04, TC-07, TC-08, TC-10, TC-14, TC-16	DEF-01, DEF-02, DEF-03, DEF-05, DEF-06, DEF-07, DEF-10	DEF-01, DEF-02, DEF-03, DEF-05, DEF-06, DEF-07, DEF-10

Member	Test cases	Defects found	Defects fixed
Fahim Hossain (0432320005101007)	TC-03, TC-05, TC-06, TC-09, TC-11, TC-12, TC-13, TC-15	DEF-04, DEF-08, DEF-09, DEF-11, DEF-12, DEF-13	DEF-04, DEF-08, DEF-09, DEF-11, DEF-12, DEF-13

13. Sign-off and Declaration

We declare that this report describes testing done on the stated version. Results can be reproduced from the repository with the commands in section 2.2. Tool use is disclosed in section 10.

Role	Name	Signature	Date
Prepared by	Md. Shakhawat Hossain, Fahim Hossain	Md. Shakhawat Hossain, Fahim Hossain	31 October 2026
Reviewed by	Al Imtiaz, Jobair Ahammed		

Appendix A: Evidence Index

All evidence files located in repository root and test output directories:

#	Evidence	File
A1	Concurrency test run results	server/test/concurrency.test.ts
A2	Server test suite execution and coverage log	vitest run (server test suite)
A3	Client build and bundle analysis	client/dist/ (tsc && vite build)
A4	Accessibility audit reports	WCAG 2.2 AA automated AST scan
A5	Documentation generation scripts	scripts/generate_all_docs.py

Appendix B: Pre-Submission Checklist

Yes: Version stated and all results come from it.

Yes: Tests run from documented commands on a fresh database.

Yes: Every in-scope requirement has a corresponding test.

Yes: Coverage numbers match the automated suite execution logs.

Yes: Zero open S1 critical defects; every fixed defect has a regression test.

Yes: Tool use disclosed transparently in section 10.

Yes: No real personal data or secret credentials in tests.

Yes: Contribution table and signatures fully documented.